Division in $\mathbb{N}$

1. $48 \div 6$
2. $144 \div 12$
3. $225 \div 15$
4. $364 \div 7$
5. $1,248 \div 24$
6. $3,150 \div 45$
7. $(48 \times 15) \div 12$
8. $(720 \div 8) \div 5$
9. A bakery boxes up 576 cookies in total, placing exactly 24 cookies into each box. How many boxes of cookies are packed?
10. A distributor receives a bulk shipment of $4,320$ items. The items are packed equally across 18 pallets, and each pallet contains 12 identical storage crates. How many items are inside each individual crate?

Solutions

1. **Answer:** 8

$$6\overline{)48}\quad 8$$

Explanation: Basic single-digit division fact: $48 \div 6 = 8$.

2. **Answer:** 12

$$12\overline{)14^{2}4}\quad 12$$

Explanation: 12 goes into 14 once with a remainder of 2, placing a 2 before the 4 to make 24, and $24 \div 12 = 2$.

3. **Answer:** 15

$$15\overline{)22^{7}5}\quad 15$$

Explanation: 15 goes into 22 once with a remainder of 7, placing a 7 before the 5 to make 75, and $75 \div 15 = 5$.

4. **Answer:** 52

$$7\overline{)36^{1}4}\quad 52$$

Explanation: 7 goes into 36 five times (35) with a remainder of 1, placing a 1 before the 4 to make 14, and $14 \div 7 = 2$.

5. **Answer:** 52

$$24\overline{)1,2^{12}4^{4}8}\quad 52$$

Explanation: 24 into 12 is 0 (remainder 12); 24 into 124 is 5 (120) with a remainder of 4; 24 into 48 is 2.

6. **Answer:** 70

$$45\overline{)3,1^{31}50}\quad 70$$

Explanation: 45 into 315 is 7 exactly (315), leaving a remainder of 0, and 45 into 0 is 0.

7. **Answer:** 60

Explanation: Evaluate using strategic grouping to avoid large numbers: first compute $48 \div 12 = 4$, then multiply by 15 to get $4 \times 15 = 60$.

8. **Answer:** 18

Explanation: Evaluate sequentially from left to right: first compute $720 \div 8 = 90$, then divide the result by 5: $90 \div 5 = 18$.

9. **Answer:** 24 **boxes**

$$24\overline{)57^{9}6}\quad\text{(quotient: 24)}$$

Explanation: 24 goes into 57 twice (48) with a remainder of 9, placing a 9 before the 6 to make 96, and $96 \div 24 = 4$.

10. **Answer:** 20 **items**

$$18\overline{)4,3^{7}20}\quad\text{(quotient: 240)}\qquad\text{then}\qquad 12\overline{)240}\quad\text{(quotient: 20)}$$

Explanation: First, find the total items per pallet ($4,320 \div 18 = 240$), carrying non-zero remainders across place values. Next, divide by crates per pallet: $240 \div 12 = 20$ items per crate.